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Set 10: Survey Responses Analysis

Data Visualization using R

Aim

To analyse survey responses using a grouped bar chart, a stacked bar chart, a table and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the survey data as a data frame.
3. Question 1: draw the grouped bar chart of Question 1 answers with `geom_bar(position = "dodge")`.
4. Question 2: draw the stacked bar chart of all three questions with `geom_bar()`.
5. Question 3: make the table with `tableGrob()`.
6. Question 4: join the two charts and the table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
survey <- data.frame(Survey_ID = 1:3, Question1 = c("A","B","C"),
                     Question2 = c("B","A","A"), Question3 = c("C","D","B"))
print(survey)
long <- stack(survey[, 2:4]); names(long) <- c("Answer", "Question")
print(table(long$Question, long$Answer))

# 1. Grouped bar chart - Question 1
p1 <- ggplot(survey, aes(Question1, fill = factor(Survey_ID))) +
  geom_bar(position = "dodge") +
  labs(title = "Answers of Question 1", x = "Answer", y = "Count", fill = "Survey ID")

# 2. Stacked bar chart - all questions
p2 <- ggplot(long, aes(Question, fill = Answer)) + geom_bar() +
  labs(title = "Responses of All Questions", x = "Question", y = "Count")

# 3. Table
tb <- tableGrob(survey, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Survey Dashboard")

ggsave("set10_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set10_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set10_q3.png", arrangeGrob(tb, top = "Survey Response Table"), width = 5, height = 2, bg = "white")
ggsave("set10_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

Survey ID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

	A	B	C	D
Question1	1	1	1	0
Question2	2	1	0	0
Question3	0	1	1	1

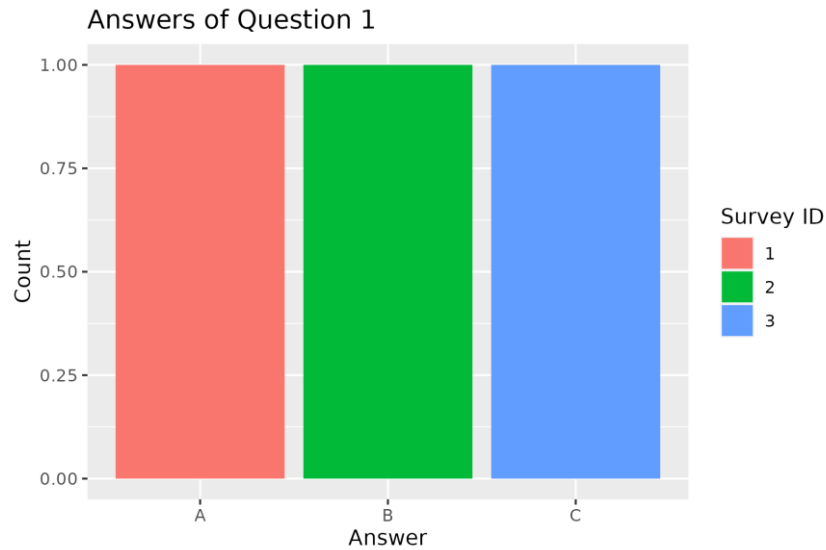


Figure 10.1: Q1 - Grouped bar chart (Question 1)

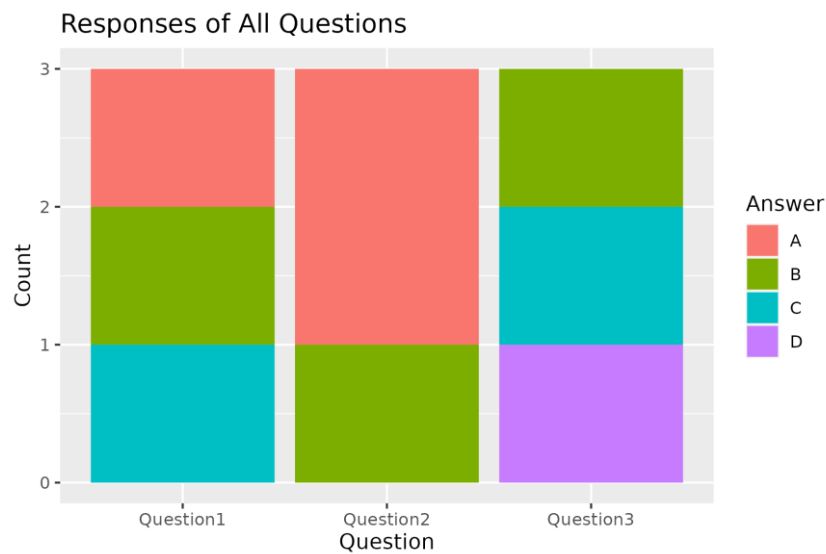


Figure 10.2: Q2 - Stacked bar chart (all questions)

Survey Response Table

Survey_ID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 10.3: Q3 - Survey response table

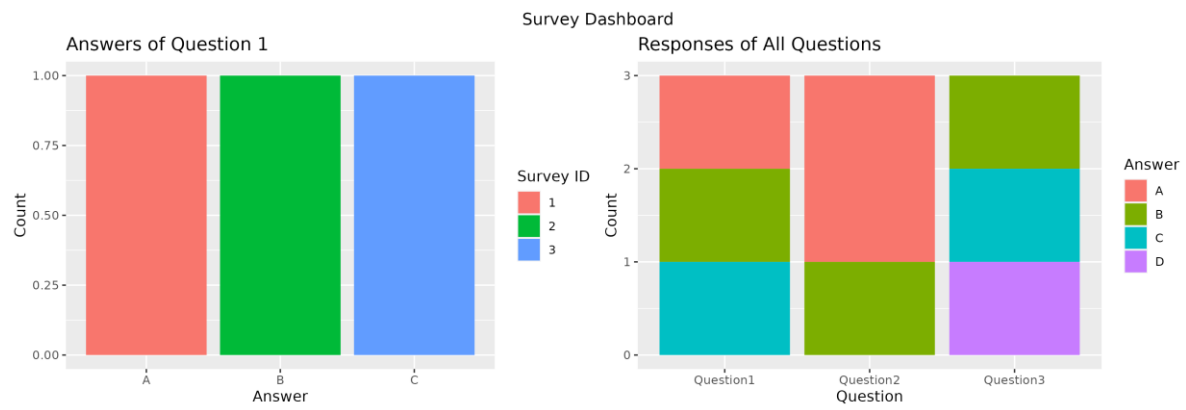


Figure 10.4: Q4 - Dashboard

Result

- In Question 1, answers A, B and C were chosen once each.
- Question 2 has A twice and B once; Question 3 has B, C and D once each.
- All four parts were completed successfully in R.

Set 11: Product Category Analysis

Data Visualization using R

Aim

To analyse product category sales using a funnel chart, a table, a pie chart and a dashboard in R.

Procedure

8. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
9. Create the sales data as a data frame and sort it from highest to lowest sales.
10. Question 1: draw the funnel chart with `geom_tile()`; the stages are the categories ranked by sales.
11. Question 2: make the table with `tableGrob()`.
12. Question 4: draw the pie chart with `geom_col()` and `coord_polar("y")`.
13. Question 3: join the pie chart, funnel chart and table into a dashboard with `arrangeGrob()`.
14. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
sales <- data.frame(Category = c("Electronics", "Clothing", "Appliances"),
                    Sales = c(50000, 35000, 40000))
sales$Percent <- round(sales$Sales / sum(sales$Sales) * 100, 1)
sales <- sales[order(-sales$Sales), ]
print(sales)
sales$Category <- factor(sales$Category, levels = rev(sales$Category))

# 1. Funnel chart
p1 <- ggplot(sales, aes(0, Category)) +
  geom_tile(aes(width = Sales, fill = Category), height = 0.8) +
  geom_text(aes(label = paste0(Category, ": $", Sales))) +
  labs(title = "Sales Funnel by Category") + theme_void()

# 2. Table
tb <- tableGrob(sales, rows = NULL)

# 4. Pie chart
p4 <- ggplot(sales, aes("", Sales, fill = Category)) + geom_col() + coord_polar("y") +
  geom_text(aes(label = paste0(Percent, "%")), position = position_stack(vjust = 0.5)) +
  labs(title = "Sales Distribution") + theme_void()

# 3. Dashboard
dash <- arrangeGrob(p4, p1, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Product Category Dashboard")

ggsave("set11_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set11_q2.png", arrangeGrob(tb, top = "Sales by Category"), width = 5, height = 2, bg = "white")
ggsave("set11_q3.png", dash, width = 12, height = 8, bg = "white")
ggsave("set11_q4.png", p4, width = 6, height = 4, bg = "white")
```

Output

	Category	Sales	Percent
1	Electronics	50000	40
3	Appliances	40000	32
2	Clothing	35000	28

Sales Funnel by Category

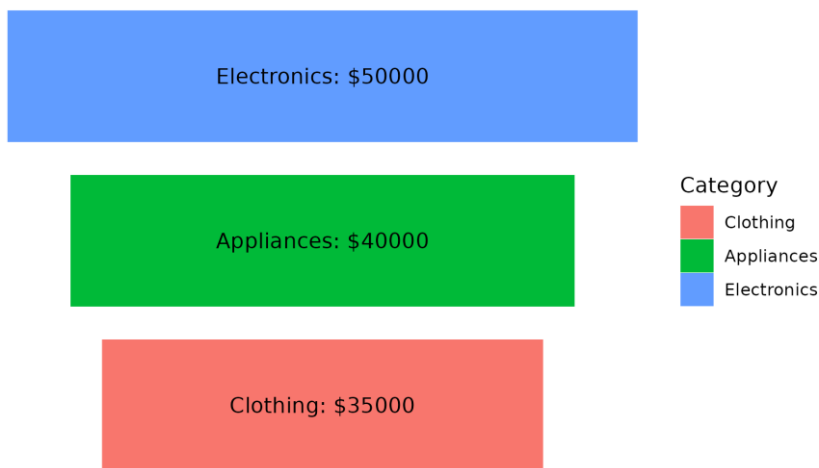


Figure 11.1: Q1 - Funnel chart

Sales by Category

Category	Sales	Percent
Electronics	50000	40
Appliances	40000	32
Clothing	35000	28

Figure 11.2: Q2 - Sales table

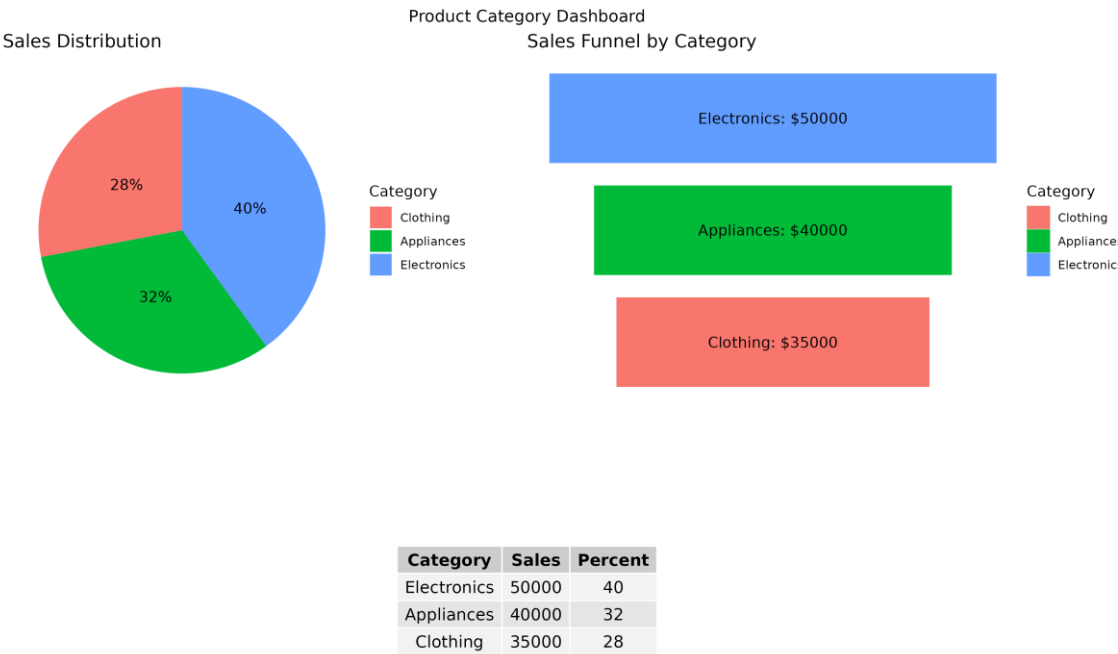


Figure 11.3: Q3 - Dashboard

Sales Distribution

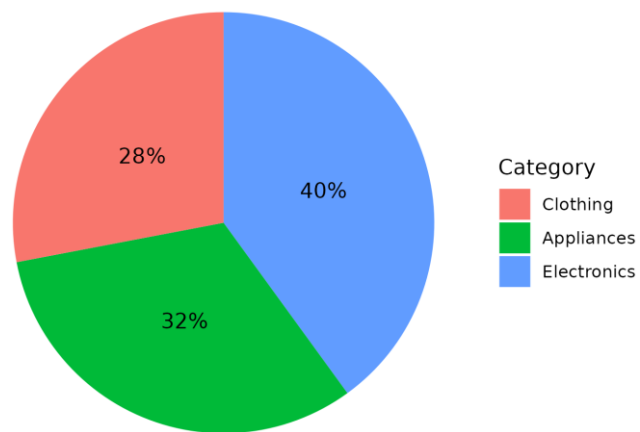


Figure 11.4: Q4 - Pie chart

Result

- Total sales are \$125,000.
- Electronics has the highest share (40%), then Appliances (32%) and Clothing (28%).
- The data has no conversion stages, so the funnel shows categories ranked by sales.

Set 12: Website Traffic Analysis

Data Visualization using R

Aim

To analyse website traffic using a line chart, a bar chart, a table and a dashboard in R.

Procedure

15. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
16. Create the traffic data as a data frame.
17. Question 1: draw the line chart of daily page views with `geom_line()`.
18. Question 2: sort by CTR, take the top N days and draw the bar chart with `geom_col()`.
19. Question 3: make the table with `tableGrob()`.
20. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
21. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
traffic <- data.frame(Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
                      Page_Views = c(1500, 1600, 1400), CTR = c(2.3, 2.7, 2.0))
print(traffic)

# 1. Line chart - daily page views
p1 <- ggplot(traffic, aes(Date, Page_Views)) + geom_line() + geom_point(size = 3) +
  labs(title = "Daily Page Views", x = "Date", y = "Page Views")

# 2. Bar chart - top N days by CTR
N <- 3
top <- head(traffic[order(-traffic$CTR), ], N)
print(top)
p2 <- ggplot(top, aes(factor(Date), CTR)) + geom_col(fill = "orange") +
  geom_text(aes(label = paste0(CTR, "%")), vjust = -0.5) +
  labs(title = paste("Top", N, "Days by CTR"), x = "Date", y = "Click-through Rate (%)")

# 3. Table
tb <- tableGrob(traffic, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Website Traffic Dashboard")

ggsave("set12_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set12_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set12_q3.png", arrangeGrob(tb, top = "Website Traffic Table"), width = 5, height = 2, bg = "white")
ggsave("set12_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

```
  Date Page_Views CTR
1 2023-01-01    1500 2.3
2 2023-01-02    1600 2.7
3 2023-01-03    1400 2.0
  Date Page_Views CTR
2 2023-01-02    1600 2.7
1 2023-01-01    1500 2.3
3 2023-01-03    1400 2.0
```

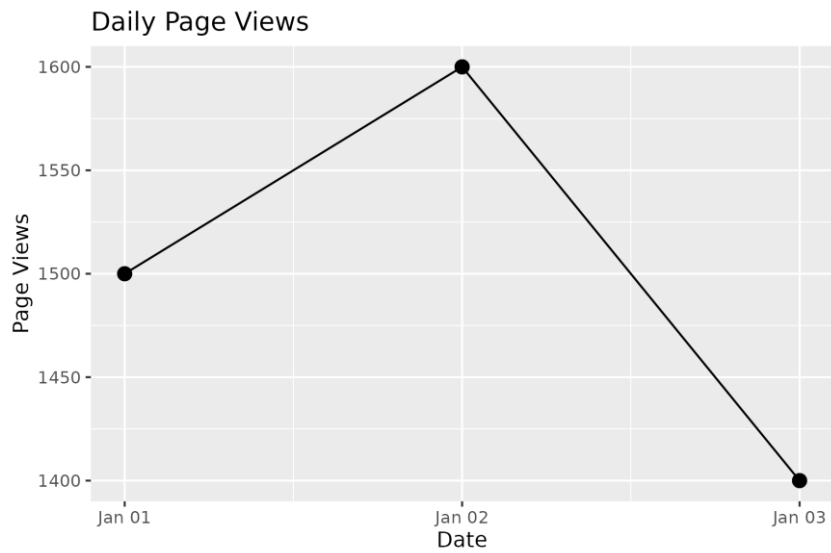


Figure 12.1: Q1 - Line chart of page views

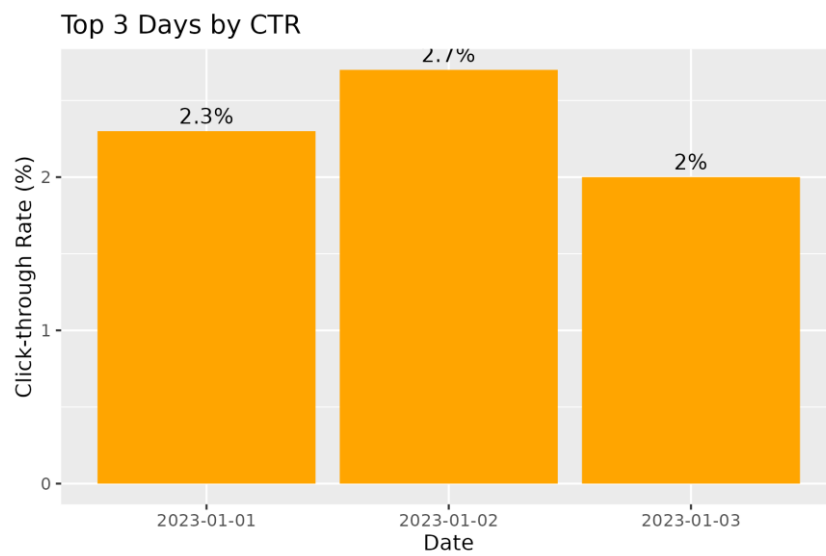


Figure 12.2: Q2 - Top N days by CTR

Website Traffic Table

Date	Page_Views	CTR
2023-01-01	1500	2.3
2023-01-02	1600	2.7
2023-01-03	1400	2.0

Figure 12.3: Q3 - Website traffic table

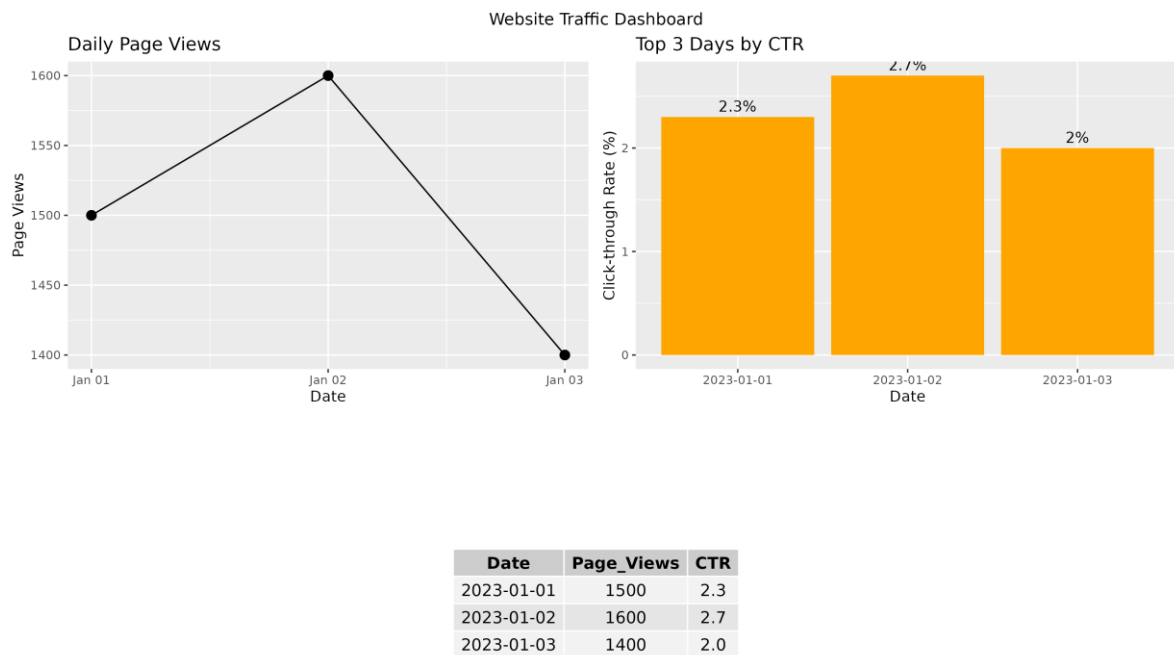


Figure 12.4: Q4 - Dashboard

Result

- Page views were highest on 2 Jan (1600) and lowest on 3 Jan (1400).
- The highest click-through rate was 2.7% on 2 Jan.
- All four parts were completed successfully in R.

Set 13: Geographic Data Analysis

Data Visualization using R

Aim

To analyse city data using a map, a scatter plot, a table and a dashboard in R.

Procedure

22. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
23. Load ggplot2, gridExtra and maps, and create the city data. The data has no coordinates, so latitude and longitude are assumed.
24. Question 1: draw the map with borders("state") and geom_point().
25. Question 2: draw the scatter plot of temperature against population and find the correlation.
26. Question 3: make the table with tableGrob().
27. Question 4: join the map, scatter plot and table into a dashboard with arrangeGrob().
28. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra); library(maps)
options(scipen = 999)

# Dataset (Lat/Lon are assumed, as the data has no coordinates)
cities <- data.frame(City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temp = c(75, 68, 80), Elevation = c(1000, 800, 1200),
  Lat = c(39.7, 41.9, 33.4), Lon = c(-105, -87.6, -112))
print(cities)

# 1. Map chart
p1 <- ggplot(cities, aes(Lon, Lat)) + borders("state") +
  geom_point(aes(size = Population), colour = "red") +
  geom_text(aes(label = City), vjust = -1.5) + coord_quickmap() +
  labs(title = "City Locations", x = "Longitude", y = "Latitude")

# 2. Scatter plot - temperature vs population
cat("Correlation:", cor(cities$Avg_Temp, cities$Population), "\n")
p2 <- ggplot(cities, aes(Avg_Temp, Population)) + geom_point(size = 4) +
  geom_text(aes(label = City), vjust = -1) + geom_smooth(method = "lm", se = FALSE) +
  ylim(450000, 750000) +
  labs(title = "Temperature vs Population", x = "Average Temperature", y = "Population")

# 3. Table
tb <- tableGrob(cities[, 1:4], rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Geographic Dashboard")

ggsave("set13_q1.png", p1, width = 7, height = 4.5, bg = "white")
ggsave("set13_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set13_q3.png", arrangeGrob(tb, top = "Geographic Data Table"), width = 6, height = 2, bg = "white")
ggsave("set13_q4.png", dash, width = 13, height = 8, bg = "white")
```

Output

	City	Population	Avg_Temp	Elevation	Lat	Lon
1	City A	500000	75	1000	39.7	-105.0
2	City B	700000	68	800	41.9	-87.6
3	City C	600000	80	1200	33.4	-112.0

Correlation: -0.5806513

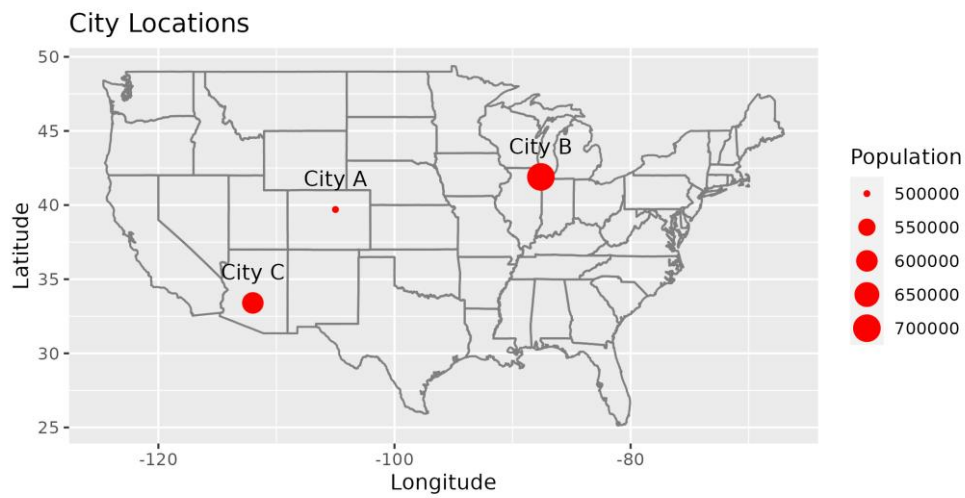


Figure 13.1: Q1 - Map of cities (coordinates assumed)

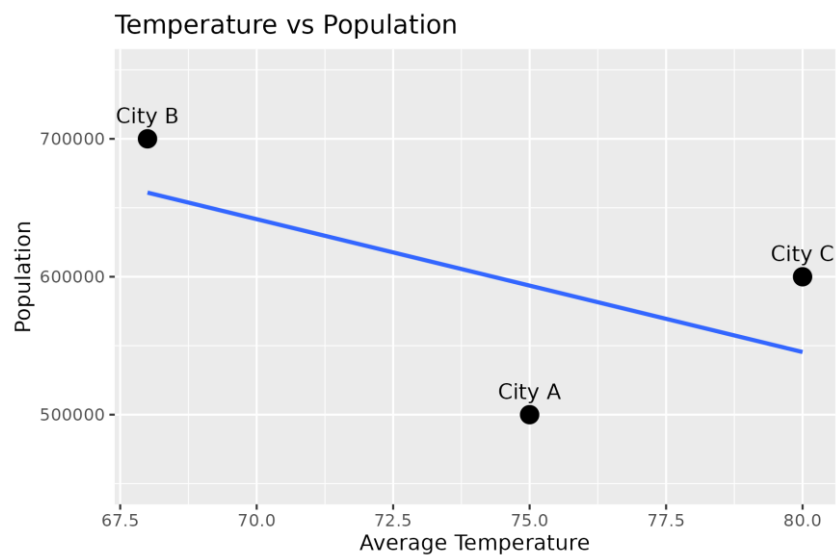


Figure 13.2: Q2 - Temperature vs population

Geographic Data Table

City	Population	Avg_Temp	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Figure 13.3: Q3 - Geographic data table

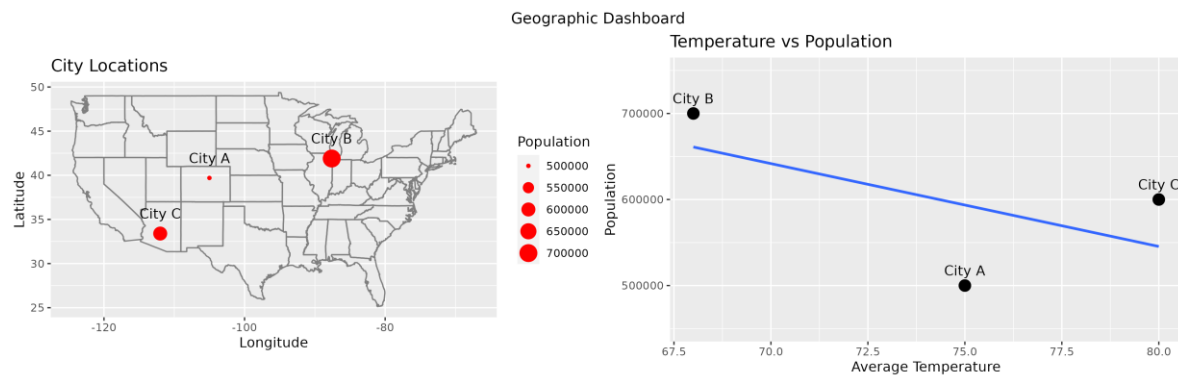


Figure 13.4: Q4 - Dashboard

Result

- Insight: the correlation is -0.58, so higher temperature goes with lower population here (City B is coolest and largest).
- With only 3 cities this is not statistically reliable.
- All four parts were completed successfully in R.